Name:_____ Period:_____

UNIT 3:

EARTH IN THE SOLAR SYSTEM

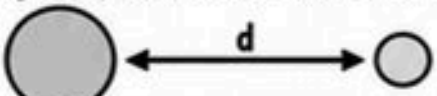
ANCHOR CHARTS



GRAVITY AND ORBITS

1 Gravity

- A **force** of attraction between any two objects with mass.
- More mass means stronger gravity.
- More distance means weaker gravity. If the distance doubles, gravity drops to ¼ (one-fourth).



More mass (stronger gravity) — **More distance** (weaker gravity)

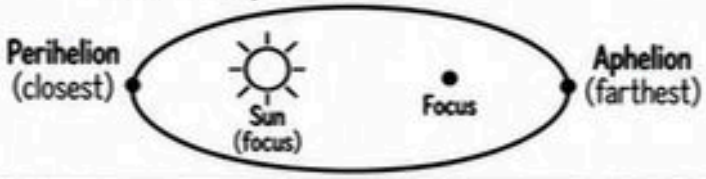
2 Orbits

- An **orbit** is the curved path of an object around a larger object.
- It happens because of a balance between two things:
  - **Inertia**: the object's forward motion (it keeps moving straight).
  - **Gravity**: pulls the object inward toward the larger body.
- The object is constantly "falling around" the larger body.



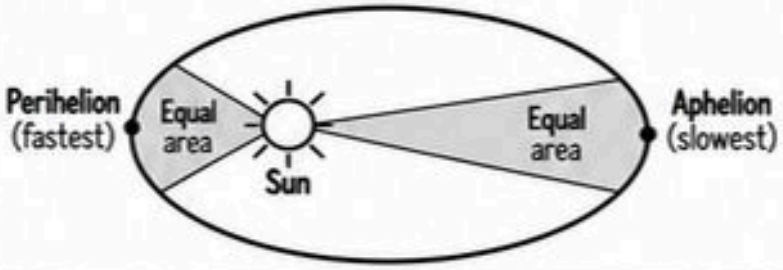
3 Kepler's 1st Law

- Orbits are **ellipses**, with the Sun at one focus.
- **Eccentricity** measures how flat the ellipse is.
  - 0 = a circle
  - Values closer to 1 = a flatter ellipse
- Earth's orbit is nearly circular (eccentricity ≈ 0.017).



4 Kepler's 2nd Law

- A planet sweeps out **equal areas** in **equal times**.
- It moves **fastest** at perihelion (closest to the Sun) and slowest at **aphelion** (farthest from the Sun).
- Gravitational force is **greatest** at perihelion.



5 Kepler's 3rd Law

- The farther a **planet** is from the Sun, the longer its **period** and the **slower** its orbital speed.

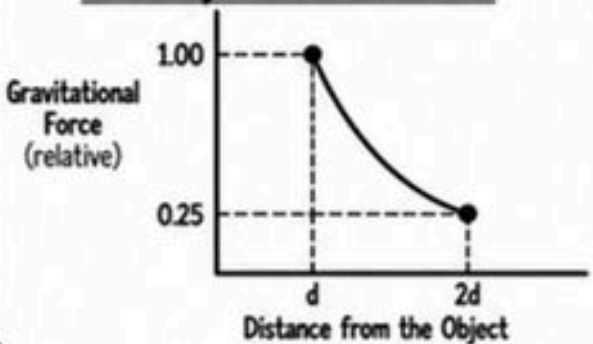
Planet	Average Distance (AU)	Period (years)
Mercury	0.39	0.24
Earth	1.00	1.00
Jupiter	5.20	11.86

Pattern:
As distance from the Sun increases, the period gets longer and orbital speed gets slower.

6 Changing Orbits and Other Key Ideas

- **Orbits can change** due to the gravity of other objects or collisions.
- **Gravity assists** use a planet's gravity to change a spacecraft's speed and direction.
- **Artemis II** uses a free-return trajectory, which uses the Moon's gravity to loop around and return to Earth without needing extra fuel.
- These laws apply to everything in orbit, including **moons** and human-made satellites, not just planets.

Gravity and Distance



Vocabulary

- **gravity** – a force of attraction between objects with mass
- **inertia** – the tendency to keep moving in a straight line
- **orbit** – the curved path of an object around another object
- **ellipse** – an oval-shaped curve
- **focus** – one of two fixed points in an ellipse
- **eccentricity** – a number that describes how flat an ellipse is
- **perihelion** – the point in an orbit closest to the Sun
- **aphelion** – the point in an orbit farthest from the Sun
- **period** – the time to complete one full orbit

⚠ Watch For:

"There's no gravity in space."

Astronauts float because they are in free fall, not because gravity is gone.

GRAVITY AND ORBITS

Part A: Fill in the Blank

1. Gravity is a force of attraction between any two objects with _____.
2. If the distance between two objects doubles, gravity drops to _____ of its original strength.
3. An orbit happens because of a balance between gravity and _____, the object's forward motion.
4. Kepler's 1st Law states that orbits are _____, with the Sun at one focus.
5. Kepler's 2nd Law states that a planet sweeps out equal _____ in equal times.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Perihelion		A. Earth takes 1.00 year to travel once around the Sun.
2. Aphelion		B. The point where a planet is farthest from the Sun and moving slowest.
3. Eccentricity		C. A spacecraft uses a planet's gravity to change its speed and direction.
4. Period		D. Earth's orbit has a value of about 0.017, so it is nearly circular.
5. Gravity assist		E. The point where a planet is closest to the Sun and gravitational force is greatest.

Part C: True / False

1. _____ More mass means weaker gravity.
2. _____ A planet moves fastest at perihelion.
3. _____ Kepler's laws apply to moons and human-made satellites, not just planets.
4. _____ Astronauts float because there is no gravity in space.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. Mars is about 1.52 AU from the Sun. Using the Kepler's 3rd Law table, predict how Mars's period and orbital speed compare to Earth's. Explain.

Period: longer / shorter Orbital speed: faster / slower

_____

_____

2. A satellite moves to twice its original distance from Earth. Use the Gravity and Distance graph to describe how the gravitational force on it changes.

_____

_____

3. Using the terms inertia and gravity, explain why the Moon does not fly off into space or crash into Earth.

_____

_____

SEASONS AND MOTION IN THE SKY

1 Earth's Motions

- **Rotation**
  - Earth spins west to east once every 24 hours.
  - Causes day and night.
  - Makes the Sun, Moon, and stars appear to move east to west at 15° per hour.



Rotation (24 hours)

- **Revolution**
  - Earth orbits the Sun once every 365¼ days.
  - Different constellations are visible in different seasons.



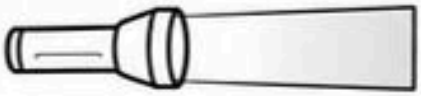
Revolution (365¼ days)

2 Tilt and Seasons

- Earth's **axis** is tilted **23.5°** and always points toward Polaris (it stays parallel to itself).
- **Seasons** are caused by **tilt** combined with **revolution**, which changes:
  - the angle of incoming sunlight (direct rays are more concentrated)
  - the length of daylight.
- **Distance does NOT** cause seasons. Earth is closest to the Sun in early January (perihelion).



3 Sunlight Angle and Intensity



Direct rays (more concentrated) = more energy (warmer)

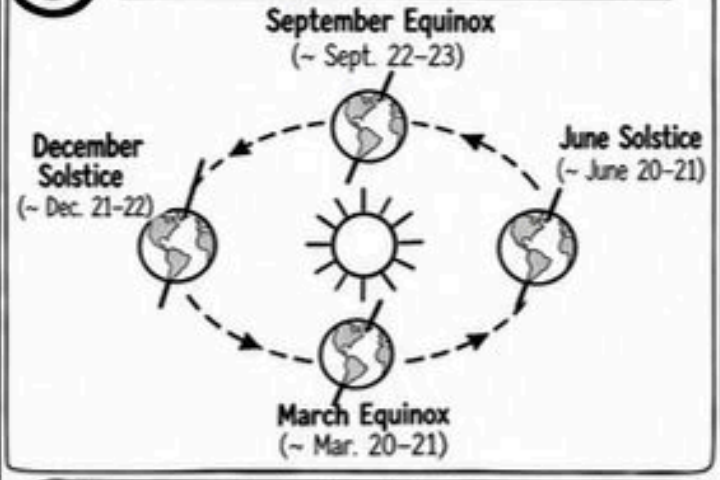


Slanted rays (more spread out) = less energy (cooler)

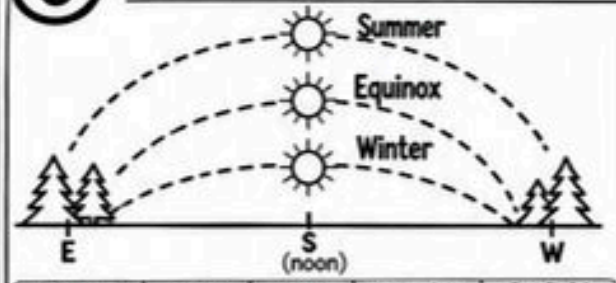
4 Solstices and Equinoxes

Event	Approx. Date	Directly Overhead At
June Solstice (summer)	June 20–21	Tropic of Cancer (23.5° N)
September Equinox (fall)	Sept. 22–23	Equator (0°)
December Solstice (winter)	Dec. 21–22	Tropic of Capricorn (23.5° S)
March Equinox (spring)	Mar. 20–21	Equator (0°)

5 Earth's Orbit and the Seasons



6 Sun's Path in New York (≈ 43° N)



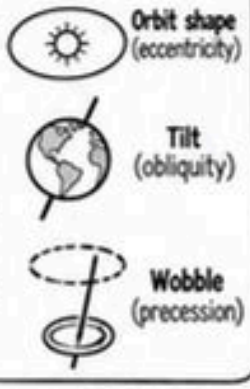
Season	Rises	Sets	Noon Sun	Daylight
Summer	NE	NW	Highest	Longest (≈ 15 hrs)
Equinox	Due E	Due W	Middle	12 hrs
Winter	SE	SW	Lowest	Shortest (≈ 9 hrs)

- The noon Sun is always due south in NY and is never directly overhead.
- **Polaris's** altitude equals your latitude, about 43° in Kenmore.

7 Long-Term Cycles

- Over tens to hundreds of thousands of years, changes in:
  - **orbit** shape (eccentricity)
  - **tilt** (obliquity)
  - **wobble** (precession)

  change how sunlight is distributed on Earth.
- These cycles are linked to ice ages and long-term climate change.



Vocabulary

- **rotation** – Earth spinning on its axis
- **revolution** – Earth orbiting the Sun
- **axis** – an imaginary line through Earth's poles
- **tilt** – the 23.5° angle of Earth's axis
- **insolation** – the amount of solar energy reaching Earth's surface
- **solstice** – when a hemisphere is tilted most toward or away from the Sun
- **equinox** – when both hemispheres get about equal daylight (≈ 12 hrs)
- **altitude** – the angle of an object above the horizon
- **Polaris** – the North Star, near Earth's axis
- **perihelion** – closest point in Earth's orbit to the Sun
- **aphelion** – farthest point in Earth's orbit from the Sun

⚠ Watch For:

- "Summer happens because we're closer to the Sun." (False)
- "The Sun is directly overhead at noon." (False in New York.)
- "The Sun always rises due east." (False. It only rises due east at the equinoxes.)

SEASONS AND MOTION IN THE SKY

Part A: Fill in the Blank

1. Earth's axis is tilted _____ degrees and always points toward Polaris.
2. Rotation makes the Sun, Moon, and stars appear to move east to west at _____ degrees per hour.
3. Seasons are caused by tilt combined with _____.
4. _____ is the amount of solar energy reaching Earth's surface.
5. Over tens to hundreds of thousands of years, changes in orbit shape, tilt, and wobble are linked to _____ and long-term climate change.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Rotation		A. Different constellations are visible in different seasons.
2. Revolution		B. The Sun is directly overhead at the Tropic of Capricorn (23.5° S).
3. June Solstice		C. Day and night happen every 24 hours.
4. December Solstice		D. The Sun rises due east, sets due west, and New York gets about 12 hours of daylight.
5. Equinox		E. The Sun is directly overhead at the Tropic of Cancer and New York has its longest daylight.

Part C: True / False

1. _____ Summer happens because Earth is closest to the Sun.
2. _____ Direct rays are more concentrated and deliver more energy than slanted rays.
3. _____ In New York, the noon Sun is directly overhead in the summer.
4. _____ The Sun always rises due east.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. It is December 21 in Buffalo, NY. Describe where the Sun rises and sets, how high the noon Sun is, and how much daylight there is.

   Rises: _____ Sets: _____ Relative Height of Sun: highest / middle / lowest Daylight: about ____ hours

2. A student says, "It's colder in winter because we're farther from the Sun." Use evidence from the chart to explain why the student is wrong.

_____

_____

3. A star rises in the east at 9 PM. How many degrees will it appear to move across the sky by 1 AM? What causes this motion?

   Degrees moved: _____

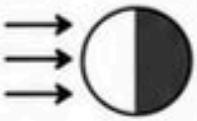
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MOON PHASES, ECLIPSES, AND TIDES

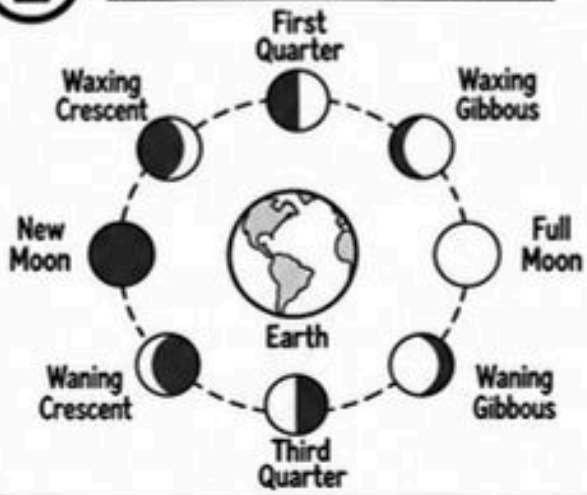
1 Moon Basics

- Half of the Moon is always lit by the Sun. The **phase** is how much of that lit half we can see from Earth.
- The full phase cycle takes about 29.5 days.
- The Moon rotates once for each revolution, so we always see the same side.
- The Moon rises about 50 minutes later each day.



The Moon is always half lit by the Sun.

2 The Phases of the Moon



Order of Phases (29.5 days)

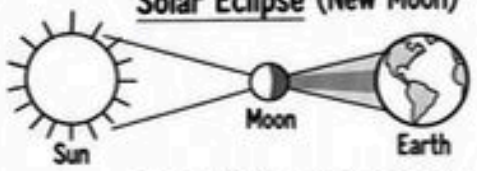
1. New Moon
2. Waxing Crescent
3. First Quarter
4. Waxing Gibbous
5. Full Moon
6. Waning Gibbous
7. Third Quarter
8. Waning Crescent

Waxing means the lit side is on the right (as seen from the Northern Hemisphere).

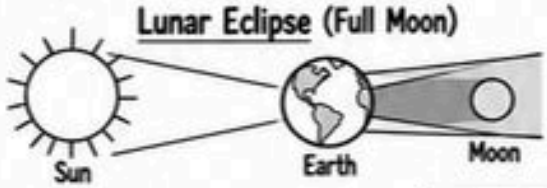
3 Eclipses

Eclipse	Moon Phase	What Is in the Middle?	Shadow	Who Can See It?
Solar Eclipse	New Moon	Moon (between Sun and Earth)	Moon's shadow falls on Earth	Only people in a narrow path (path of totality)
Lunar Eclipse	Full Moon	Earth (between Sun and Moon)	Earth's shadow falls on the Moon	The whole night side of Earth

Solar Eclipse (New Moon)



Lunar Eclipse (Full Moon)



- **Umbra**: the full, darkest part of a shadow.
- **Penumbra**: the lighter, partial part of a shadow.
- Eclipses don't happen every month because the Moon's orbit is tilted about 5° compared to Earth's orbit.

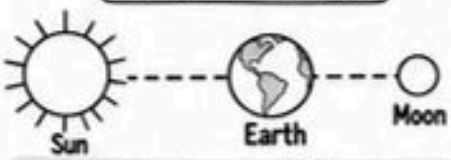


Local Connection: Buffalo was in the path of totality on April 8, 2024.

4 Tides

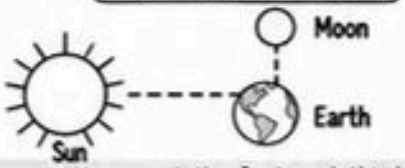
- Tides are caused mainly by the Moon's gravity, with help from the Sun.
- Most coasts get two high tides and two low tides about every 24 hours and 50 minutes.
- High tides are about 12 hours and 25 minutes apart.

Spring Tides (largest tidal range)



Happen at new and full moon when the Sun, Earth, and Moon line up.

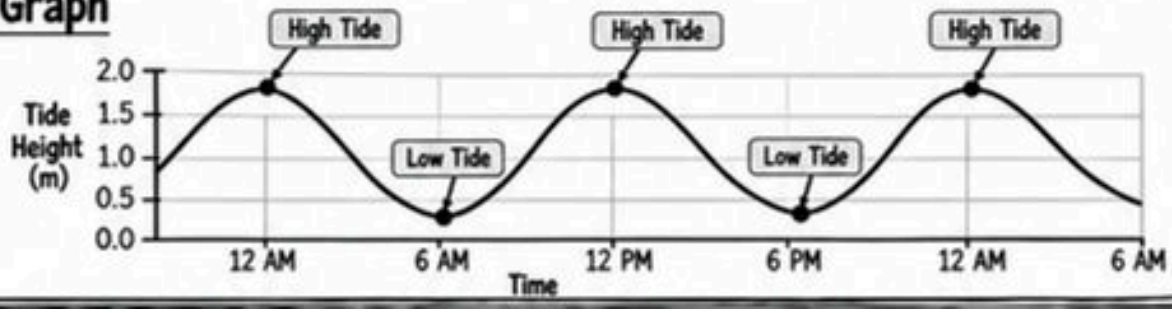
Neap Tides (smallest tidal range)



Happen at the first and third quarter moon when the Sun and Moon are at 90° to each other.

5 Tide Height Graph

- Shows the repeating pattern of tides.
- About two high tides and two low tides in ~24 hr 50 min.
- High tides are ~12 hr 25 min apart.



MOON PHASES, ECLIPSES, AND TIDES

Part A: Fill in the Blank

1. Half of the Moon is always lit by the _____.
2. The full phase cycle of the Moon takes about _____ days.
3. The _____ is the full, darkest part of a shadow.
4. Eclipses don't happen every month because the Moon's orbit is tilted about _____ degrees compared to Earth's orbit.
5. Tides are caused mainly by the Moon's _____, with help from the Sun.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Solar eclipse		A. Earth's shadow falls on the Moon, and the whole night side of Earth can see it.
2. Lunar eclipse		B. The Sun and Moon are at 90° to each other at first quarter, giving the smallest tidal range.
3. Spring tides		C. A thin sliver of the Moon is lit on the right side, the phase right after new moon.
4. Neap tides		D. The Sun, Earth, and Moon line up at full moon, giving the largest tidal range.
5. Waxing crescent		E. The Moon's shadow falls on Earth, and only people in a narrow path see totality.

Part C: True / False

1. _____ The Moon's phases happen because Earth's shadow covers part of the Moon.
2. _____ We always see the same side of the Moon because it rotates once for each revolution.
3. _____ A solar eclipse can only happen during a full moon.
4. _____ The Moon rises about 50 minutes earlier each day.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. Buffalo was in the path of totality on April 8, 2024. What was the Moon's phase, and what was in the middle? Why did people outside the narrow path not see totality?

   Moon phase: _____ Object in the middle: _____

2. Use the tide height graph. A high tide happens at 12:00 PM. Predict the time of the next high tide and explain the pattern.

   Next high tide: _____

3. Tonight is a full moon. Will a coastal town have spring tides or neap tides? Explain using the positions of the Sun, Earth, and Moon.

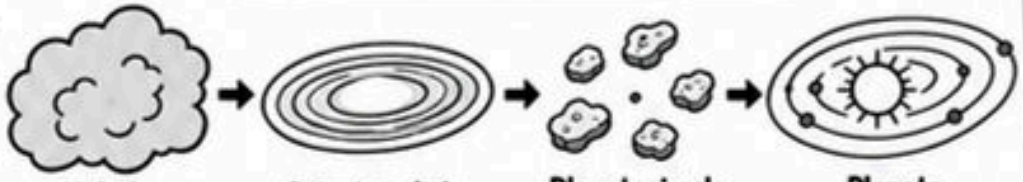
   Type of tide: _____

SOLAR SYSTEM FORMATION AND EARLY EARTH

1 The Nebular Theory

- About **4.6 billion years** ago, a cloud of gas and dust (nebula) collapsed under gravity.
- It spun and flattened into a disk.
- The Sun formed at the center.
- The planets formed by **accretion** (particles sticking together to make larger bodies).

From a Nebula to a Solar System



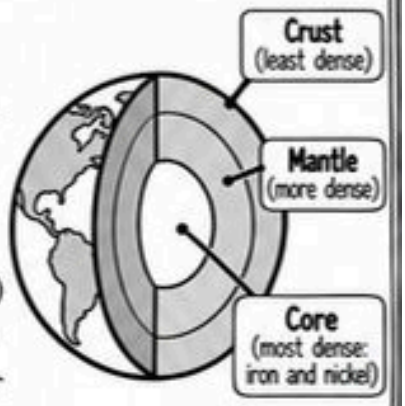
Nebula (cloud of gas and dust) → **Spinning disk** (flat, rotating disk of material) → **Planetesimals** (small bodies that collide and grow) → **Planets** (larger bodies that form the solar system)

2 Evidence for the Nebular Theory

Evidence	What it tells us
Planets orbit in the same direction and in nearly the same plane.	Formed from a single, spinning disk.
The inner planets are rocky and the outer planets are gas and ice giants.	Material was sorted by distance from the Sun (hotter near, cooler far).
Telescopes have observed dusty disks around young stars.	Shows that planet formation is happening around other stars today.

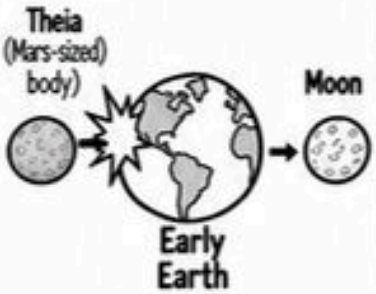
3 Early Earth

- Early Earth was **molten** from frequent impacts and heat.
- Material separated by **density**: iron and **nickel** sank to form the core, with the mantle and crust above (differentiation).
- Gases released by volcanic outgassing (H_2O vapor, CO_2, N_2) formed the early atmosphere, which had no free oxygen.
- Oceans formed as water vapor cooled and condensed.



4 Moon Formation

- The Moon most likely formed from a giant impact between early Earth and a Mars-sized body (Theia).
- The impact sent rock into orbit around Earth, which came together to form the Moon.



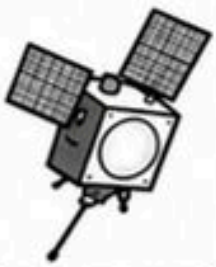
5 How We Know the Age

- **Radiometric dating** uses radioactive decay, which happens at a constant rate (half-life), to determine age.
- **Meteorites** date to about **4.56 billion years**.
- **Moon rocks** date to about **4.4–4.5 billion years**.
- Earth's oldest minerals (**zircon crystals**) date to about **4.4 billion years**.



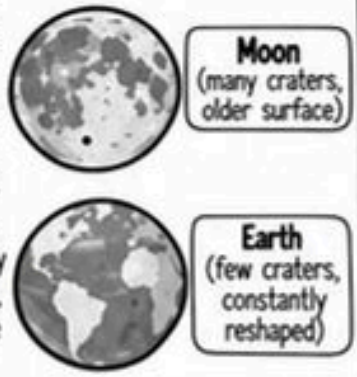
6 Why Earth's Oldest Rocks Are Younger

- Plate tectonics, weathering, erosion, and the rock cycle have destroyed or changed most of Earth's early rocks.
- Meteorites, asteroids, and the Moon have changed very little, so we study them to learn about early Earth.
- Example: **OSIRIS-REx** collected samples from the Bennu, which help us study the original material from the early solar system.



7 Impact Craters as Evidence

- **More craters** means an older surface (it has been exposed for a longer time).
- The Moon's heavily cratered highlands are older than its smoother maria (lava plains).
- Earth has few visible craters because its surface is constantly being reshaped by weathering, erosion, volcanism, and plate tectonics.



SOLAR SYSTEM FORMATION & EARLY EARTH

Part A: Fill in the Blank

1. About _____ billion years ago, a cloud of gas and dust (nebula) collapsed under gravity.
2. The planets formed by _____, particles sticking together to make larger bodies.
3. In early Earth, iron and nickel sank to form the _____.
4. Radiometric dating uses radioactive decay, which happens at a constant rate called _____.
5. The Moon most likely formed from a giant impact between early Earth and a Mars-sized body called _____.

Part B: Matching

Match each term to its example.

Term	Answer	Example
1. Nebula		A. Volcanoes released H2O vapor, CO2, and N2, forming Earth's early atmosphere.
2. Planetesimals		B. Earth's oldest minerals, dated to about 4.4 billion years.
3. Differentiation		C. Material separated by density, with the densest material sinking to the center.
4. Outgassing		D. A cloud of gas and dust that collapsed under gravity.
5. Zircon crystals		E. Small bodies that collided and grew into planets.

Part C: True / False

1. _____ The inner planets are rocky because material was sorted by distance from the Sun.
2. _____ Earth's early atmosphere had a lot of free oxygen.
3. _____ The Moon's smooth maria are older than its heavily cratered highlands.
4. _____ Earth has few visible craters because its surface is constantly being reshaped.

Part D: Short Answer / Application

Use what you know, along with the anchor chart, to answer the following.

1. A rock sample is dated to 4.56 billion years, older than any rock on Earth. What object did it most likely come from? Why do scientists study these objects to learn about early Earth?

   Object: _____

   _____

   _____

2. A telescope observes a dusty disk spinning around a young star. Explain how this observation supports the nebular theory.

   _____

   _____

3. Surface A has many craters. Surface B has very few. Which surface is older? Explain using the chart.

   Older surface: _____

   _____

   _____